

Concrete Technology

BRANCH: CIVIL ENGG. (CE)

COURSE CODE: CE301

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Concrete Technology (CE301) under the Civil Engg. (CE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Cement & Aggregates

Manufacturing of Portland cement, chemical composition, hydration of cement, Bogue's compounds, physical testing. Aggregates: classification, sizing, gradation, bulking of sand.

Unit 2: Fresh & Hardened Concrete

Workability, factors affecting workability, testing: Slump test, Compacting Factor, Vee-Bee. Hardened concrete: compressive, tensile, flexural strength, stress-strain behavior. Creep, shrinkage.

Unit 3: Concrete Admixtures

Chemical admixtures: plasticizers, superplasticizers, retarders, accelerators, air-entraining agents. Mineral admixtures: Fly ash, Silica fume, GGBS.

Unit 4: Mix Design & Special Concrete

IS Code method of Concrete Mix Design (IS 10262). Quality control. Special concretes: High Strength Concrete, Self-Compacting Concrete, Fiber-Reinforced Concrete.

*Slump values for different placements, target mean strength: $f'_{ck} = f_{ck} + 1.65 * s$ (where s = std deviation).*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



